

Mid-Point Summary

Phases A–B: landscape, competitive intelligence, telemetry analytics, and anomaly detection

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KEY DECISION TODAY

Select the default anomaly-detection path for Rover / Sentinel-style fleet health monitoring.

10

seeded train/test
splits

150

controlled fault
events

0.985

overall LOF AUROC

Phase A–B review frame

MID-POINT REVIEW

A four-week path from ecosystem orientation to statistically tested anomaly recommendation

Week 1	Landscape + toolchain	Week 2	Competitive intelligence	Week 3	Telemetry analytics	Week 4	Anomaly benchmark
	Physical AI brief, PIC 2.0 map, Python environment	Vendor matrix, three strategic gaps, InGen crosswalk		Rover generator, EDA, feature dictionary		Fault injection, 4-method test, recommendation memo	

WHAT HAS BEEN DELIVERED

Foundation	5-reading landscape synthesis; PIC 2.0 conceptual map; reproducible Python 3.11 toolchain.	Data system	Seedable Rover telemetry generator plus EDA and 25-feature dictionary.
Market view	Competitive matrix across humanoid, service, and patrol vendors; gap memo tied to InGen platforms.	Evidence base	Controlled fault injection; 4-method benchmark; Wilcoxon-tested recommendation.

Mid-point question: which anomaly-detection method should become the default path for Phases C–D integration?

Phase A: strategic grounding

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Weeks 1–2 established the product context and competitor baseline for the technical benchmark

CORE CONCLUSION

Physical AI is moving toward a platform contest, not a single-robot contest.

That shifted the analysis from “which robot is best?” to “which intelligence, telemetry, safety, and learning layers can compound across products?”

THREE STRATEGIC GAPS CARRIED INTO PHASE B

1 Fragmented competitor field

Most vendors emphasize one body or one vertical; InGen can frame “one intelligence layer, many embodiments.”

2 Under-served human-centered adoption

Care, education, and companion use cases need trust, safety, and decision quality, not only robot hardware.

3 Security needs governance-aware anomaly logic

Patrol/security systems require uncertainty, escalation discipline, and auditable false-alert control.

PRODUCT ANCHOR LOGIC

Origami / PIC 2.0

Foundation layer for uncertainty, safety, adaptive control, and fleet learning.

Aido Rover + Sentinel

Best Phase B substrate: telemetry-rich, operations-facing, and naturally tied to anomaly detection.

Implication: use Week 4 to test whether fleet-health monitoring can be made statistically defensible, not just plausible.

Week 3: telemetry foundation

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A reproducible Rover dataset and feature layer created the substrate for the Week 4 benchmark

432K

synthetic rows in default
EDA sample

5

Rover units in
sample

129.6M

generator full-scale
capacity

25

documented
features

1 Hz

sample rate

Generator coverage

- joint / actuator angles
- IMU accel + gyro
- motor current per drive
- battery voltage + SoC
- WiFi RSSI
- GPS fix quality
- task-success flags

Feature engineering logic

Short-window statistics

rolling means, slopes, RMS
vibration, missingness rate

Cross-sensor ratios

voltage vs SoC, speed vs
current, GPS vs WiFi
consistency

Fleet-relative z-scores

battery, motor current, RSSI
compared across units

Why it matters for Week 4

The benchmark compares detectors on engineered signals that carry physical meaning, not only raw sensor values.

Week 4 benchmark design

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Controlled fault injection with leakage controls and paired statistical testing

FAULT PATTERNS INJECTED INTO TEST BLOCKS ONLY

Motor stall	speed collapse + one drive current spike + thermal rise	GPS jitter	spoofing-like position jumps + HDOP instability
Sensor drift	gradual IMU accel / gyro calibration shift	Software hang	stale telemetry blocks + task-success impact
Battery degradation	accelerated SoC drain + voltage sag		

EVALUATION DISCIPLINE

10 seeded splits	same split seeds across all four methods	Clean train windows	faults injected after split and only in held-out blocks
Scaler fit on train only	no test-window statistics leak into normalization	Paired Wilcoxon tests	AUROC differences compared split-by-split

Design choice:

evaluate one-minute windows rather than isolated seconds, reflecting how an operations dashboard would triage fleet health.

Benchmark results: LOF leads accuracy

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Overall metrics across five fault types and 10 seeded train/test splits

Overall AUROC and F1

	AUROC		F1	
Isolation Forest		0.844		0.365
One-Class SVM		0.981		0.617
LOF		0.985		0.651
LSTM AE		0.848		0.466

Accuracy evidence

LOF beat Isolation Forest by +0.141 AUROC and LSTM AE by +0.137 AUROC; both paired Wilcoxon $p = 0.002$. LOF also beat One-Class SVM by +0.003 AUROC with $p = 0.0195$, but the operational gap is small.

Per-fault readout

LOF ranked best on 4/5 faults. One-Class SVM was marginally best on software hang and effectively tied with LOF.

Recommendation by operating regime

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The preferred method changes depending on latency, accuracy, and escalation risk

Latency-constrained edge screen

Use Isolation Forest as a first-pass screen only, paired with explicit guards for battery, GPS, stale telemetry, and task-success degradation.

Caveat: benchmark AUROC was 0.844 ± 0.044 , materially below LOF and One-Class SVM.

Accuracy-maximizing analytics

Use LOF as the default model for offline review, fleet-health triage, and analyst dashboards.

Caveat: neighbor-based scoring may become heavier at 50-unit, 30-day scale and needs stress testing.

High-stakes escalation

Use LOF plus rule-based safety checks; keep One-Class SVM as the challenger for software-hang-like faults.

Caveat: thresholds must be calibrated to severity, false-alert tolerance, and operator workload.

Explicit limitations

- Synthetic telemetry and controlled faults, not real field failures.
- One-minute windows do not measure sub-second alert latency.

- Fixed contamination threshold should be recalibrated before field use.
- Re-run needed on the full 50-unit, 30-day sample before final deployment claims.

Bottom line:

adopt LOF as the primary accuracy path, keep One-Class SVM as challenger, and use Isolation Forest only as a lightweight screen with guardrails.

Next phase: convert analysis into policy and dashboard decisions

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Phases C–D will test decision behavior and package the results for stakeholders

What is ready now

- Clean repo structure and Week 1–4 artifacts
- A seedable Rover telemetry generator
- EDA and feature dictionary aligned to fleet health
- Benchmark pipeline with leakage controls and significance testing
- Recommendation memo with operating-regime caveats

Phases C–D focus

- Week 5** PPO / SAC benchmarks and multi-agent coordination lessons
- Week 6** AMDC-style decision-evaluation harness with reliability analysis
- Week 7** Streamlit dashboard: fleet health, policy benchmark, decision scorecards
- Week 8** Capstone report and executive deck

Mid-point message:

Phases A–B produced a defensible evidence chain from landscape framing to statistically tested anomaly -method selection.